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Hiroki Sukeno

Research Profile My research explores quantum many-body physics through the lens of entanglement as an organizing principle. I investigate how the interplay of entanglement and measurement enables new approaches to quantum simulation, including measurement-based and adaptive quantum circuits. My work integrates ideas from field theory, statistical mechanics, and quantum information, with connections to experimental AMO platforms. Recent efforts focus on novel entanglement structures, their dynamical behavior, and measurement-driven protocols for simulating complex quantum systems.

Positions

The University of New Mexico — Albuquerque, NM, USA
Postdoctoral Fellow — June 2025 — present
Funded by NSF FRHTP Program

Education

Stony Brook University — Stony Brook, NY, USA
C. N. Yang Institute for Theoretical Physics — Ph.D. degree
Ph.D. candidate (advisor: Tzu-Chieh Wei) — May 2025
M.A. degree in May 2020

The University of Tokyo — Meguro, Tokyo, Japan
Master of Arts (advisor: Yuji Okawa) — M.A. degree
- High Energy Theory — March 2017

The University of Tokyo — Bunkyo, Tokyo, Japan
Department of Physics — B.S. degree
- Focus on Astrophysics Experiment — March 2015

Publications

(I) Measurement-based frameworks for gauge symmetry and dynamics

1. **H. Sukeno**, K. Ikeda, and T.-C. Wei, "*Bulk and boundary entanglement transitions in the projective gauge-Higgs model*," Phys. Rev. B **110**, 245102 (2024). *[First author]*
2. T. Okuda, A. Parayil Mana, and **H. Sukeno**, "*Anomaly inflow for CSS and fractonic lattice models and dualities via cluster state measurement*," SciPost Physics **17**, 113 (2024). *[Co-first author (alphabetically ordered)]*
3. T. Okuda, A. Parayil Mana, and **H. Sukeno**, "*Anomaly inflow, dualities, and quantum simulation of abelian lattice gauge theories induced by measurements*," Phys. Rev. Research **6**, 043018 (2024). *[Co-first author (alphabetically ordered)]*
4. **H. Sukeno** and T.-C. Wei, "*Quantum simulation of lattice gauge theories via deterministic duality transformations assisted by measurements*," Phys. Rev. A **109**, 042611 (2024). *[First author]*
5. **H. Sukeno** and T. Okuda, "*Measurement-based quantum simulation of Abelian lattice gauge theories*," SciPost Phys. **14**, 129 (2023). *[First author]*

(II) Topological phases of matter and generalized symmetries via quantum circuits

6. A. Parayil Mana, Y. Li, **H. Sukeno**, and T.-C. Wei, *"Higher-order topological phases protected by non-invertible and subsystem symmetries"* cond-mat/2505.18119
7. A. Parayil Mana, Y. Li, **H. Sukeno**, and T.-C. Wei, "*Kennedy-Tasaki transformation and non-invertible symmetry in lattice models beyond one dimension*," Phys. Rev. B **109**, 245129 (2024).
8. Y. Li, **H. Sukeno**, A. Parayil Mana, H. P. Nautrup, T.-C. Wei, "*Symmetry-enriched topological order from partially gauging symmetry-protected topologically ordered states assisted by measurements*," Phys. Rev. B **108** (11) 115144 (2023).

(III) Quantum teleportation protocols & quantum algorithms

9. **H. Sukeno** and T.-C. Wei, "*Quantum gate broadcasting on graphs*" quant-ph/2503.10946 *[First author]*
10. N. A. Nghiem, **H. Sukeno**, S. Zhang, and T.-C. Wei, "*Improved Quantum Power Method and Numerical Integration Using a Quantum Singular-Value Transformation*," Phys. Rev. A **111**, 012434 (2025).
11. R. K. Malla, **H. Sukeno**, H. Yu, T.-C. Wei, A. Weichselbaum, and R.M. Konik, "*Feedback-based Quantum Algorithm Inspired by Counterdiabatic Driving*," Phys. Rev. Research 6, 043068 (2024).
12. **H. Sukeno**, T.-C. Wei, M. Hillery, J. Bergou, D. Fields, and V. S. Malinovski, "*Broadcasting single-qubit and multi-qubit-entangled states: authentication, cryptography, and distributed quantum computation*," Phys. Rev. A **107** (6) 062605 (2023). *[First author]*

(IV) Problems in broader high energy & condensed matter physics

13. S. Zhang, **H. Sukeno**, K. Ikeda, and T.-C. Wei, "*Local symmetries and extensive ground-state degeneracy of a one-dimensional supersymmetric fermionic chain*," Phys. Rev. B **111**, 235151 (2025).

14. H. Kunitomo, Y. Okawa, **H. Sukeno**, and T. Takezaki, "*Fermion scattering amplitudes from gauge-invariant actions for open superstring field theory*," arXiv preprint hep-th/1612.00777 *[Alphabetically ordered]*

TALK, SEMINAR, AND LECTURE

1. [**seminar** ***(invited)***] "Lattice gauge theory and adaptive quantum circuits," Seminar at the High Energy Theory group of the University of Tokyo, February 2026.
2. [**conference**] "Quantum gate broadcasting on graphs"APS Physics Global Summit, Anaheim, March 2025.
3. [**seminar** ***(postdoc interview)***] "Physics and quantum simulation of lattice gauge theory with mid-circuit measurement," Seminar at University of New Mexico, January 2025.
4. [**seminar** ***(postdoc interview)***] "Physics and quantum simulation of lattice gauge theory with mid-circuit measurement," Seminar at University of California Davis, January 2025.
5. [**poster**] "Bulk and boundary entanglement transitions in the projective gauge-Higgs model," Quantum information dynamics and non-equilibrium quantum matter, Simons Center for Geometry and Physics, Stony Brook University, December 2024.
6. [**seminar** ***(postdoc interview)***] "Lattice gauge theory from entanglement, measurement, and feedforward," Special Seminar at Pritzker School of Molecular Engineering in The University of Chicago, November 2024.
7. [**seminar** ***(postdoc interview)***] "Topological orders, quantum simulation, and quantum communication — physics in mid-circuit measurement paradigm ," Seminar at Virginia Tech (Blacksburgh), November 2024.
8. [**seminar** ***(postdoc interview)***] "Lattice gauge theory from entanglement, measurement, and feedforward," PCTS Special Seminar at Princeton University, October 2024.
9. [**seminar** ***(postdoc interview)***] "Lattice gauge theory from entanglement, measurement, and feedforward," GLAM Special Seminar at Stanford University, October 2024.
10. [**seminar**] "Lattice gauge theory from entanglement, measurement, and feedforward," YITP Stony Brook University, October 2024.
11. [**seminar**] "Lattice gauge theory from entanglement, measurement, and feedforward," High Energy Theory group at Osaka University, October 2024.
12. [**poster**] "Bulk and boundary entanglement transitions in the projective gauge-Higgs model," Physics of Quantum Information 2024, Perimeter Institute for Theoretical Physics, Waterloo, Canada
13. [**conference**] "Anomaly inflow, foliation, and measurement-based quantum simulation of abelian lattice gauge theories," APS March Meeting 2024, Minneapolis, USA
14. [**lecture** ***(invited)***] "Measurement-based quantum computation and lattice gauge theory," school held at Osaka University, Japan, October 2023

15. [**conference**] "Lattice gauge theories from measuring entangled states," Foundations and Developments of Quantum Information Theory, Yukawa Institute for Theoretical Physics, Kyoto University, Japan, September 2023
16. [**conference**] "Measurement-based quantum simulation of Abelian lattice gauge theories," It from Qubit 2023, Perimeter Institute, Canada, August 2023
17. [**seminar** ***(invited)***] "Measurement-based quantum simulation of Abelian lattice gauge theories," Hybrid RBRC seminar, Brookhaven National Laboratory, USA, May 2023
18. [**seminar** ***(invited)***] "Quantum Simulation of Gauge Theories from Entanglement, Measurement, and Feedforward," Extreme Universe Collaboration Circular Meeting, Japan, online, April 2023
19. [**seminar** ***(invited)***] "Quantum Simulation of Gauge Theories from Measuring Entangled States," Joint HEP-TH Seminar, China, online, March 2023
20. [**conference**] "Fermion scattering amplitudes from gauge-invariant actions of open superstring field theory," String Field Theory and String Phenomenology 2018, Harish-Chandra Research Institute, India, February 2018
21. [**seminar** ***(invited)***] "Fermion scattering amplitudes from gauge-invariant actions for open superstring field theory," seminar at Tokyo Institute of Technology, Japan, 2017
22. [**poster**] "Fermion scattering amplitudes from gauge-invariant actions for open superstring field theory," YITP international workshop Strings and Fields 2016, Yukawa Institute for Theoretical Physics, Japan, 2016
23. [**conference**] "Fermion scattering amplitudes from gauge-invariant actions of open superstring field theory," Japan Physics Society 2016, Tohoku Gakuin University, Japan, 2016

FELLOWSHIP

- Focused Research Hubs in Theoretical Physics Prize Fellow, 2025 – present
- Ito Foundation U.S.A.-FUTI scholarship, 2018-2020: a scholarship for international graduate studies

AWARD

- Rosaline and Milton Sterman Travel Award, YITP Stony Brook University, August 2024
- Peter B. Kahn Prize, outstanding research and travel award, Stony Brook University, May 2023
- Outstanding Student Award, School of Arts and Sciences, University of Tokyo, March 2017
- Full repayment exemption from JASSO student loan for academic excellence, March 2017

REFEREE SERVICE

- Conference: Quantum Information Processing 2025, 2026
- Journal
  APS: Physical Review A, Physical Review X, PRX Quantum, Physical Review Letters

Nature Publishing group: Communications Physics
Springer Nature: International Journal of Theoretical Physics

ACADEMIC SERVICE

- Co-organizer of SQuInT 2025
- Volunteer program facilitator for QIS 303, Quantum Error Mitigation Program, C2QA, August 2024
- Member of Extreme Universe Collaboration, Grant-in-Aid for Transformative Research Areas (A)

MENTORING SERVICE

- Mentored a Master student of Prof. Wei at Stony Brook (see preprint 2206.14028).
- Volunteer TA for Quantum EduQation Professional Development Workshop, Stony Brook University, 2023-2024.
- Summer school advisor for General Incorporated Association Glocal Academy, August 2018.

TEACHING POSITIONS

- The University of Tokyo. Teaching Assistant: Waves and Oscillations, First-Year Seminar for Natural Sciences Students.
- Stony Brook University. Teaching Assistant: Physics Lab (undergraduate), Graduate Lab, Physics Lab for Life Sciences.
- Waseda Academy Co. Part-time lecturer: high school math & physics (2013-2018)

LINK

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